INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY BANGALORE

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Department of Computer Science & Information Technology

DS 601: Applied Machine Learning

MAJOR PROJECT ASSIGNMENT

Customer Lifetime Value Intelligence Engine

Probabilistic Forecasting, Uplift Modelling & Marketing ROI Optimizer

Course Code	DS 601	**Semester**	II (Even Semester)
Academic Year	2024–25	**Credits**	4
Instructor	Prof. Rajesh Sundaresan	**Designation**	Professor & Area Chair
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Issue Date	**20 January 2025**	**Due Date**	**28 March 2025, 11:59 PM IST**
Max Marks	**100 + 20 Bonus**	**Project Type**	Individual / Group of 2

Academic Integrity Notice

All submitted work must be your own. You may use open-source libraries and public datasets, but all code must be written by you. Use of AI-generated code that you do not understand and cannot explain will be treated as a violation of the IIIT-B Academic Integrity Policy. Any plagiarism – including between student groups – will result in an **F grade** and referral to the Disciplinary Committee. You are required to submit a signed *Academic Integrity Declaration Form* along with your final submission.

§1 Project Overview & Motivation

Modern businesses generate vast quantities of transactional data, yet most data science teams reduce the entire customer value problem to a binary question: *will this customer churn?* This is intellectually and commercially insufficient. The correct question is: **how much revenue will each customer generate over their lifetime, and how should we allocate finite marketing resources to maximise that revenue?**

This project asks you to build a production-grade **Customer Lifetime Value (CLV) Intelligence Engine** – an end-to-end machine learning system that predicts individual customer CLV using a hybrid of classical probabilistic models and modern gradient-boosted trees, segments customers into actionable tiers, estimates the causal uplift of marketing interventions, and optimises budget allocation to maximise expected incremental revenue. A fully interactive Streamlit dashboard must expose all of these capabilities to a non-technical business user.

This problem is directly industrially relevant: companies such as Amazon, Flipkart, Swiggy, Zepto, and Nykaa make CLV-driven decisions daily. Upon completing this project, you will have demonstrated mastery of probabilistic modelling, ensemble learning, causal ML (uplift), uncertainty quantification, and applied business optimisation – a portfolio that spans the full ML value chain.

Learning Objectives

Upon completing this project, students will be able to:

LO1. Derive and implement the BG/NBD and Gamma-Gamma probabilistic models from first principles and interpret their parameters in a business context.

LO2. Engineer RFM and behavioural feature sets from raw transactional data and perform rigorous EDA including cohort survival analysis.

LO3. Train, tune, and stack gradient-boosted ensemble models (XGBoost / LightGBM) for regression tasks with proper temporal cross-validation.

LO4. Apply SHAP-based global and instance-level explainability to ML regression models.

LO5. Implement the T-Learner meta-learner for uplift modelling and evaluate it using Qini curves.

LO6. Formulate and solve a constrained marketing budget allocation problem using linear programming.

LO7. Produce calibrated probabilistic predictions using conformal prediction intervals.

LO8. Monitor model health using Population Stability Index (PSI) and detect concept drift.

LO9. Deploy a multi-tab Streamlit dashboard with business-facing interactive analytics.

§2 Dataset Requirements

2.1 Primary Dataset MANDATORY

You must use the **UCI Online Retail II Dataset** (`https://archive.ics.uci.edu/dataset/502/online+retail+ii`). This dataset contains 1,067,371 transactions from a UK-based online retailer spanning December 2009 to December 2011. It includes invoice number, stock code, description, quantity, invoice date, unit price, customer ID, and country.

Data Quality Warning

The dataset contains approximately 24% rows with missing Customer IDs, negative quantities (representing returns and cancellations), zero-price entries, and duplicate invoice lines. **Your preprocessing pipeline must handle all of these cases explicitly and document every decision.** Dropping rows without justification will be penalised.

2.2 Supplementary / Extension Dataset BONUS

For the bonus extension, you may additionally use the **Olist Brazilian E-Commerce Dataset** (`https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce`), which contains richer metadata including product category, seller, review scores, and delivery timelines. Using both datasets and presenting a comparative CLV analysis earns up to 10 bonus marks.

2.3 Temporal Split Protocol MANDATORY

Split	Period	Purpose	Notes
Observation Window	Dec 2009 – Nov 2010	Feature engineering, model training	12 months of behaviour
Holdout Window	Dec 2010 – Feb 2011	CLV label ground truth	3-month forward revenue
Monitoring Slices	Quarterly sub-splits	PSI / drift analysis	Q1, Q2, Q3, Q4

No data from the holdout window may be used during training or feature construction. Leakage will be detected during the viva and will result in a zero for the modelling component.

§3 Required System Architecture

Your system must implement all six layers described below. Each layer is independently evaluated.

Layer 1: Data Ingestion, Cleaning & RFM Feature Engineering [20 marks]

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Layer 2: Probabilistic CLV Modelling (BG/NBD + Gamma-Gamma) [20 marks]

↓

Layer 3: ML Augmentation (LightGBM) + Stacking Ensemble [15 marks]

↓

Layer 4: Customer Segmentation (GMM) + SHAP Explainability [10 marks]

↓

Layer 5: Uplift Modelling (T-Learner) + Budget Optimiser (LP) [15 marks]

↓

Layer 6: Conformal Intervals + PSI Drift Monitor [10 marks]

↓

Layer 7: Streamlit Dashboard (4 Tabs: Explorer / Segments / Optimizer / Monitor) [10 marks]

§4 Detailed Technical Requirements

4.1 Layer 1 — Data Preprocessing & Feature Engineering MANDATORY

- Remove duplicate rows; handle cancellations (invoices starting with 'C') separately – do not simply drop them. Analyse return behaviour as a customer feature.
- Impute or justify removal of missing Customer IDs. Document the fraction of revenue lost.
- Construct a clean transaction table: one row per (CustomerID, InvoiceDate, OrderValue).
- Engineer the following features per customer over the observation window:
  - **RFM:** Recency (days since last purchase), Frequency (distinct invoice count), Monetary (mean and total order value, log-transformed).

- **Behavioural:** Return rate, category diversity (Shannon entropy over product categories), weekend purchase ratio, average inter-purchase gap, coefficient of variation in order value.
- **Temporal:** Customer tenure, number of active months, recency trend (is recent frequency increasing or decreasing?).

- Produce a **cohort retention heatmap** (cohort month on y-axis, months since acquisition on x-axis, retention rate as colour). This is mandatory in your report.
- Perform Winsorisation at the 99th percentile for all monetary features.

4.2 Layer 2 — Probabilistic CLV Modelling MANDATORY

You must implement and understand the following models. Use of the `lifetimes` library is permitted for fitting, but you must be able to explain the mathematical foundation in the viva.

BG/NBD Model:

- Fit on (x, t_x, T) tuples from the observation window, where x = frequency, t_x = recency, T = customer age.
- Plot: expected vs. actual transaction counts by frequency decile (calibration check).
- Compute $P(\text{alive} \mid x, t_x, T)$ for every customer.
- Analyse: which customer characteristics correlate with high dropout probability?

Gamma-Gamma Model:

- Fit on customers with $x \geq 2$ purchases (the model assumes purchase frequency and monetary value are independent – verify this assumption via a correlation test and report the result).
- Produce: expected average order value per customer.

Combined CLV:

$$\widehat{\text{CLV}}_i = \mathbb{E}[\text{transactions}_i \mid \text{alive}] \times \mathbb{E}[\text{avg. order value}_i] \times \text{gross margin}$$

Assume gross margin = 0.60. Predict 3-month forward CLV (to match holdout period).

4.3 Layer 3 — ML Augmentation & Stacking MANDATORY

- Train a **LightGBM regressor** on the feature matrix constructed in Layer 1, with the BG/NBD outputs ($P(\text{alive})$, expected transactions) as additional features.
- Target variable: actual 3-month revenue in the holdout window (revenue regression).
- Use **time-series aware cross-validation** (no random shuffling; the validation set must always be temporally after the training set).
- Tune hyperparameters using **Optuna** (minimum 50 trials).
- Build a **stacking ensemble**: train a meta-learner (Ridge regression or another LightGBM) on out-of-fold predictions from [BG/NBD, LightGBM]. The stacked model must outperform each individual model – demonstrate this with a results table.
- Report: MAE, RMSE, MAPE, and Spearman rank correlation between predicted and actual CLV.

Model Evaluation Note

For CLV, **rank ordering** (Spearman ρ) is often more business-relevant than absolute error (MAE). A model that correctly identifies your top 10% customers by CLV is extremely valuable even if its point estimates are imprecise. Report both.

4.4 Layer 4 — Customer Segmentation & Explainability MANDATORY

- Cluster customers using a **Gaussian Mixture Model** on the 3D feature space: $(\log(\widehat{\text{CLV}}), P(\text{alive}), \text{recency})$
- Use BIC to select the number of components (evaluate $k \in \{3, 4, 5, 6\}$).
- Name and operationalise each segment with a recommended business action (e.g., "Win-Back Campaign," "VIP Loyalty Programme," "Sunset").
- Produce a 2D scatter plot of $P(\text{alive})$ vs. $\log(\widehat{\text{CLV}})$ coloured by segment.
- Generate SHAP explanations:
  - **Global:** beeswarm plot of top 15 features.
  - **Group-level:** mean SHAP values per segment – which features drive CLV in each segment?
  - **Instance-level:** waterfall plots for one representative customer from each segment.

4.5 Layer 5 — Uplift Modelling & Budget Optimisation MANDATORY

Uplift Modelling (T-Learner):

- Since a real A/B experiment is unavailable, you will construct a **synthetic treatment assignment**: randomly assign 30% of customers to a "treated" group. Simulate the treatment effect as:

$$P(\text{purchase in 30 days} \mid \text{treated}) = \text{sigmoid}(0.3 \cdot \text{CLV score} + 0.15 \cdot \text{treatment})$$

  Document this simulation explicitly and discuss its limitations.
- Fit separate LightGBM classifiers on treatment and control groups; compute uplift $\hat{\tau}_i = \hat{\mu}_1(x_i) - \hat{\mu}_0(x_i)$.
- Evaluate using the **Qini curve** and **AUUC** (Area Under the Uplift Curve).
- Key insight to demonstrate: show customers with high CLV but low uplift ("always-buyers") and discuss why targeting them wastes budget.

Budget Optimisation:

- Define: cost per contact $c = £2.00$, expected incremental revenue per persuadable = £18.00.
- Formulate a **linear programme**:

$$\max_{z_i \in \{0,1\}} \sum_i z_i \cdot \hat{\tau}_i \cdot \text{revenue}_i \quad \text{subject to} \quad \sum_i z_i \cdot c \leq B$$

- Solve using `scipy.optimize.linprog` (LP relaxation) and compare with a greedy sort-by-ROI baseline.
- Plot the **ROI curve**: x-axis = budget B (£0 to £20,000), y-axis = expected incremental revenue.

- Report the optimal budget, the number of customers to target, and the projected ROI.

4.6 Layer 6 — Uncertainty Quantification & Monitoring MANDATORY

- Implement **Conformal Prediction Intervals** using the `MAPIE` library (Split Conformal, $\alpha = 0.10$, i.e., 90% coverage).
- Validate: in the holdout set, do 90% of actual CLV values fall within the predicted interval? Report the empirical coverage and average interval width.
- Implement **Population Stability Index (PSI)** for RFM features across quarterly slices:

$$\mathrm{PSI} = \sum_{j=1}^{J} \left(\mathrm{Actual}\%_j - \mathrm{Expected}\%_j\right) \ln\left(\frac{\mathrm{Actual}\%_j}{\mathrm{Expected}\%_j}\right)$$

- Apply the standard thresholds: PSI < 0.10 (stable), 0.10–0.20 (slight shift), > 0.20 (significant drift, model recalibration recommended).
- Display a warning banner in the dashboard when any feature's PSI exceeds 0.20.

4.7 Layer 7 — Streamlit Dashboard MANDATORY

The dashboard must contain exactly four tabs:

Tab	Required Elements
Tab 1: CLV Explorer	Searchable / filterable customer table with predicted CLV, confidence interval, segment, and P(alive); SHAP waterfall for any selected customer row.
Tab 2: Segment Intel	2D scatter (interactive, Plotly); segment summary cards (count, avg CLV, recommended action); mandatory cohort retention heatmap.
Tab 3: Budget Optimiser	Budget slider (£0–£20k); live ROI curve (updates on slider change); output cards: customers to target, expected incremental revenue, projected ROI multiple; scenario radio button (Conservative / Base / Optimistic ±30%).
Tab 4: Model Health	PSI bar chart per feature per quarter; CLV distribution overlay (current vs. base); model metrics summary; red warning banner when PSI > 0.20.

§5 Constraints & Restrictions

Hard Constraints — Violations Will Be Penalised

C1. No data leakage. The holdout window (Dec 2010 – Feb 2011) must not be visible to any model during training, feature engineering, or hyperparameter selection.

C2. No random shuffling for time-series CV. All cross-validation must respect temporal order.

C3. No black-box API calls for core predictions. You may not use the OpenAI API, Gemini API, or any hosted LLM to generate CLV predictions. These must come from your own trained models.

C4. Reproducibility. All random seeds must be set and reported. Your code must produce identical results on two consecutive runs.

C5. No pre-trained CLV-specific libraries (e.g., `pymc-marketing` CLV module or `btyd` wrappers that hide model internals). You must be able to explain every parameter of the BG/NBD model.

C6. Documentation. Every notebook must have markdown cells explaining each step. Uncommented code blocks of more than 10 lines will be flagged.

C7. Group constraint. Groups of two must divide work clearly in the report. Both members must be able to explain all components during the viva. Individual contributions must be stated in a CRediT author statement.

§6 Deliverables & Submission Format

Submission Checklist

Submit a single **.zip archive** named DS601_MP_<RollNumber>.zip (or DS601_MP_<Roll1>_<Roll2>.zip for groups) containing the following structure:

```
DS601_MP_<RollNumber>/
|-- report/
|   `-- DS601_MP_Report.pdf          # 15--25 pages, IEEE two-column format
|-- code/
|   |-- 01_EDA_preprocessing.ipynb
|   |-- 02_probabilistic_models.ipynb
|   |-- 03_ml_layer_stacking.ipynb
|   |-- 04_segmentation_shap.ipynb
|   |-- 05_uplift_budget_optimiser.ipynb
|   |-- 06_uncertainty_monitoring.ipynb
|   `-- app.py                       # Streamlit dashboard entry point
|-- data/
|   `-- README_data.txt              # Instructions to download dataset
|-- models/
|   `-- (saved model files: .pkl / .joblib / .cbm)
|-- requirements.txt
|-- README.md
`-- academic_integrity_declaration.pdf
```

6.1 Technical Report Requirements MANDATORY

The report must follow the **IEEE two-column conference format** and cover:

1. **Abstract** (200 words): problem, approach, key results.

2. **Introduction**: business motivation, why CLV > churn modelling (cite at least 3 papers).
3. **Related Work**: BG/NBD literature (Fader et al. 2005 is mandatory), uplift modelling survey, conformal prediction reference.
4. **Dataset & EDA**: cohort retention heatmap, feature distributions, data quality decisions.
5. **Methodology**: detailed description of all 6 layers with equations.
6. **Results**: model comparison table (Table 1), Qini curve, ROI curve, SHAP plots.
7. **Dashboard**: screenshots of all 4 tabs.
8. **Discussion**: limitations, ethical considerations (CLV-based targeting fairness), future work.
9. **Conclusion**.
10. **References** (minimum 8 academic citations).
11. **Appendix**: CRediT author statement (group projects only).

Table 1: Expected format of the Model Comparison Table (fill with your results)

Model	MAE (£)	RMSE (£)	MAPE (%)	Spearman ρ	Coverage	Interval
BG/NBD Baseline					–	–
LightGBM (standalone)					–	–
Stacked Ensemble					90% target	
Best Model						

§7 Evaluation Rubric

Component	Marks	Criteria
Data Preprocessing & EDA	20	Correctness of cleaning (5), cohort heatmap (5), feature engineering depth (5), EDA insights and visualisations (5)
Probabilistic Modelling (BG/NBD + GG)	20	Model correctness (8), calibration plots (4), CLV formula implementation (4), insight on $P(\text{alive})$ (4)
LightGBM + Stacking	15	Temporal CV (4), Optuna tuning (3), stacking implementation (4), results table (4)
Segmentation & SHAP	10	GMM + BIC (4), segment naming and business logic (3), SHAP plots (3)
Uplift + Budget Optimiser	15	T-Learner correctness (5), Qini/AUUC (4), LP formulation (3), ROI curve (3)
Conformal Intervals & PSI	10	Conformal coverage validation (5), PSI implementation (3), alerting logic (2)
Streamlit Dashboard	10	All 4 tabs present (4), interactivity (3), UI quality and usability (3)
Technical Report	10	Structure and writing quality (3), equations and methodology (4), all required figures (3)
Total	**100**	
Bonus: Olist dataset comparison	+10	Cross-dataset CLV comparison analysis
Bonus: Causal fairness analysis	+5	CLV targeting bias audit by region/category
Bonus: Deployed public demo	+5	Live Streamlit Cloud link in README

Viva Voce

A viva will be conducted in Week 10 (7–11 April 2025). Duration: 15 minutes per student (30 minutes for groups). You will be asked to explain the BG/NBD model mathematics, your temporal cross-validation design, the uplift modelling formulation, and any figure from your report. **You must be able to run your Streamlit dashboard live during the viva.** Marks may be deducted if you cannot explain components you have submitted.

§8 Recommended Project Timeline

Week	Dates	Milestones
Week 1	20–26 Jan	Dataset download; full preprocessing pipeline; cohort retention heatmap; EDA notebook complete
Week 2	27 Jan–2 Feb	BG/NBD model fit; calibration plots; $P(\text{alive})$ analysis; Gamma-Gamma fit; combined CLV computation
Week 3	3–9 Feb	LightGBM training; temporal CV; Optuna tuning; stacking ensemble; results table
Week 4	10–16 Feb	GMM segmentation; SHAP global + group + instance plots; segment business logic document
Week 5	17–23 Feb	T-Learner uplift model; Qini curve; LP budget optimiser; ROI curve
Week 6	24 Feb–2 Mar	Conformal prediction intervals (MAPIE); PSI implementation; drift alerting logic
Week 7	3–9 Mar	Streamlit dashboard (all 4 tabs); end-to-end integration testing; demo video recording
Week 8	10–28 Mar	Technical report writing; README; code cleanup; final testing; submission

Checkpoint: A mandatory mid-project check-in is scheduled during lab hours in **Week 4 (10–14 February)**. You must demonstrate: (a) a working preprocessing pipeline, (b) BG/NBD model fit with calibration plots. Groups that do not demonstrate these will receive an advisory warning.

§9 Recommended Resources

9.1 Essential Reading

- Fader, P.S., Hardie, B.G.S., & Lee, K.L. (2005). "Counting your customers" the easy way: An alternative to the Pareto/NBD model. *Marketing Science, 24*(2), 275–284. [**Must read for BG/NBD**]

- Radcliffe, N.J. (2007). Using control groups to target on predicted lift: Building and assessing uplift models. *Direct Marketing Analytics Journal*, 1(3), 14–21. [**Must read for uplift modelling**]

- Angelopoulos, A.N., & Bates, S. (2023). Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning, 16*(4), 494–591.

9.2 Libraries & Tools

Library	Version	Use
`lifetimes`	$\geq$ 0.11.3	BG/NBD and Gamma-Gamma fitting
`lightgbm`	$\geq$ 4.3.0	Gradient boosted trees
`optuna`	$\geq$ 3.6.0	Hyperparameter optimisation
`shap`	$\geq$ 0.45.0	Model explainability
`mapie`	$\geq$ 0.8.0	Conformal prediction intervals
`causalml`	$\geq$ 0.15.0	T-Learner uplift meta-learner
`scipy`	$\geq$ 1.13.0	Linear programming
`streamlit`	$\geq$ 1.35.0	Dashboard
`plotly`	$\geq$ 5.20.0	Interactive charts
`mlflow`	$\geq$ 2.12.0	Experiment tracking (recommended)

9.3 Starter Code Reference

A minimal skeleton repository will be made available on the course Moodle page by **22 January 2025**. It includes: data loading utilities, the temporal split function, a BG/NBD fitting stub, and the Streamlit app skeleton. You are not required to use it, but it establishes the expected code quality standard.

§10 Submission Instructions

1. Upload the `.zip` archive to the **DS 601 Moodle portal** under "Major Project Submission" before **28 March 2025, 11:59 PM IST**.
2. The portal will close automatically at midnight. **No extensions will be granted except for documented medical emergencies with prior notice to the instructor.**
3. Late submissions will incur a penalty of **10 marks per 24-hour period**.
4. Submissions exceeding 500 MB will be rejected. Use Git LFS for large model files or provide download instructions in `README.md`.
5. Ensure your code runs end-to-end with `python -m streamlit run app.py` after `pip install -r requirements.txt`. Code that fails to execute on the TA's machine will receive zero for the dashboard component.

Final Advice from the Instructor

The goal of this project is not to maximise a single accuracy metric. It is to build a system that a real business could use to make better decisions. Ask yourself at every stage: *"If I were the Chief Marketing Officer, what would I want to know from this output?"*
The best projects I have seen are not the ones with the highest MAPE – they are the ones where the student clearly understood the business problem, made thoughtful modelling choices, and communicated their findings with clarity and intellectual honesty. A well-reasoned discussion of why your model fails in certain cases is far more impressive than an overfit leaderboard number.
Good luck. Start early. The BG/NBD model will take more time than you expect.

— *Prof. Rajesh Sundaresan*

DS 601 Major Project · IIIT Bangalore · Semester II, AY 2024–25

For queries, contact ds601.ta@iiitb.ac.in or visit office hours.

This document supersedes all prior verbal instructions. Version 1.0, issued 20 January 2025.